Assignment 5 CSE 16-02: Applied Discrete Mathematics
Winter 2026

Instructor: Cedric Westphal
Due February 12th, 2026 11:59 PM

Complete the questions carefully and show your work. Please remember to:

- Scan the pages and combine them into a single PDF file. Make sure your work is readable and that pages are oriented correctly.
- Upload your PDF file to Canvas/Gradescope under Assignment 5.
- In Gradescope select the page(s) for each problem.

Notation: $\emptyset$ is the empty set, $\mathbb{N}$ is the set of natural numbers, $\mathbb{Z}$ is the set of integers, $\mathbb{Q}$ is the set of rational numbers, and $\mathbb{R}$ is the set of real numbers. $\mathscr{P}(A)$ is the power set of A.
[46 points total in assignment.]

0. Each answer is clearly written and each answer is correctly mapped to its rubric on Gradescope. [4 points]

1. For $a, b \in \mathbb{N}$, prove the following statements below. Hint: to prove that g is the $\gcd(a, b)$ you need to prove that $g|a$ and $g|b$ and that g is the greatest such common divisor.**[10 points]**

   (a) Let $a = 5x$ and $b = 5y + 7$ for some x and y in $\mathbb{N}$. Prove that $\gcd(a, b) = \gcd(x, b)$. Write a proof that shows: (i) that $\gcd(a, b)|x$, then (ii) show that it implies that $\gcd(a, b)$ equals $\gcd(x, b)$. **[2 points]**

   (b) If both a and b are divisible by 4, then $\gcd(a, b) = 4 \gcd(a/4, b/4)$. **[2 points]**

   (c) If m is a common divisor of a and b, show the general case of the previous question, namely that $\gcd(a, b) = m \gcd(a/m, b/m)$. **[2 points]**

   (d) Prove that $\gcd(a, b) = \gcd(a + b, b)$. **[2 points]**

   (e) Is it true or false that, for $a > b$, $2\gcd(a, b) = \gcd(a + b, a - b)$? Give a clear explanation of your reasoning. **[2 points]**

2. We will prove that if n is an odd integer $\in \mathbb{N}$, then 16 divides $n^4 - 1$. We're going to do this in two ways. **[7 points]**

   (a) First, prove that if n is an odd integer $\in \mathbb{N}$, then 8 divides $n^2 - 1$.

   (b) Prove that if n is an odd integer, then so is $n^2 + 1$.

   (c) Using the identity $n^4 - 1 = (n^2 - 1)(n^2 + 1)$ and your results from steps 1 and 2, prove that 16 divides $n^4 - 1$.

   (d) Let $n = 2k + 1$. Express $n^4 - 1$ using the binomial theorem as a sum of terms of the form $a_i k^i$. Write this sum explicitly.

   (e) Consider the last two terms of your sum, a_2k^2 and a_1k. Rewrite $a_2k^2 + a_1k$ in the form $bk^2 + c(k^2 + k)$ where b is an integer.

   (f) Show that b and $c(k^2 + k)$ is divisible by 8.

   (g) Prove the result stated at the beginning of this Problem, using the previous three steps.

3. Prove or disprove: $\forall x \in \mathbb{R}, \sqrt{x} \notin \mathbb{Q} \rightarrow x \notin \mathbb{Q}$. **[5 points]**

4. Prove or disprove: $\sqrt[3]{3} \in \mathbb{Q}$. **[5 points]**

5. Prove: If $n, m \in \mathbb{Z}$ and $7n^4 + 11m^3 = 214$ then n and m have the same parity. **[5 points]**

6. Prove or disprove: If $x \notin \mathbb{Q}$, $y \in \mathbb{R}$, $x \neq 0$, and $y \neq 0$, then for all $n, m \in \mathbb{N}$, $x^n y^m \notin \mathbb{N}$. **[5 points]**

7. Consider the sets

$$S_1 = \{k^2 + 5 : k \text{ is an even natural number}\}, S_2 = \{4k + 5 : k \in \mathbb{N}\}.$$

   (a) Prove $S_1 \subseteq S_2$. **[3 points]**

   (b) Prove $\neg(S_2 \subseteq S_1)$. **[2 points]**